Notes: Moser, Voena & Waldinger (AER, 2014)

German Jewish Émigrés and US Invention

Contents

1 Big picture

- **Question.** Did the arrival of German Jewish émigré scientists from Nazi Germany increase US invention, and if so, through which mechanism?
- **Setting.** The paper studies chemistry, where patenting is a relatively good measure of innovation, and focuses on research fields defined by 166 USPTO technology classes linked to German and Austrian academic chemists.
- **Historical shock.** The Nazi dismissal of Jewish and politically targeted academics after 1933 forced many scientists out of German and Austrian universities. Some moved to the United States.
- **Treatment definition.** A research field is exposed to émigrés if it includes at least one US patent by a German or Austrian chemist who emigrated to the United States.
- **Baseline comparison.** Changes in US patenting by domestic US inventors in émigré fields are compared with changes in fields of other German chemists.
- **Main result.** Domestic US patenting rises substantially in émigré fields after the dismissals. Baseline OLS estimates imply roughly a 31 percent increase; IV estimates using pre-1933 fields of dismissed chemists imply even larger effects.
- **Key identification move.** The paper instruments for émigré fields using pre-1933 patent fields of dismissed German and Austrian chemists. These fields were determined before the United States could attract the scientists or before the post-1933 US innovation response.
- **Mechanism.** The increase is driven mainly by entry of new US inventors into émigré fields, not by higher productivity of incumbent US inventors.
- **Network channel.** Co-inventors of émigrés and co-inventors of those co-inventors later become especially active in émigré fields, suggesting persistent knowledge transmission through research networks.
- **Economic takeaway.** High-skilled immigration can raise innovation not only by the immigrants' own output, but by attracting and training new domestic inventors into the immigrants' research fields.

2 Incremental contribution of the paper

- **First systematic historical quantification.** Historical accounts long argued that German Jewish scientists transformed US science. This paper provides a systematic patent-based estimate of that effect.
- **Field-level identification rather than location-level exposure.** Prior work on high-skilled immigration often relies on city or state exposure. This paper compares research fields, which avoids some geographic sorting concerns and fits the idea that scientific knowledge moves through research areas.
- **New immigrant-scientist and patent-field data.** The paper constructs new datasets linking German and Austrian academic chemists, dismissals, migration histories, US patents, USPTO classes, inventor identities, and co-invention networks.
- **Instrument based on forced dismissals.** The IV design uses pre-1933 fields of dismissed chemists to address endogenous field choice by émigrés. This distinguishes the effect of émigré arrival from the possibility that émigrés selected into fields already poised to grow.
- **Separates total innovation from incumbent productivity.** Existing work on scientific migration often focuses on incumbents. This paper decomposes the aggregate effect into incumbent productivity and entry, showing that entry by new domestic inventors is the main force.
- **Network propagation evidence.** The co-inventor and second-degree co-inventor analyses move beyond a simple treatment effect and identify a plausible diffusion mechanism.
- **Shows negative selection can still produce large positive spillovers.** The historical narrative suggests émigrés to the United States may not have been positively selected into the best fields. IV estimates larger than OLS imply that even negatively selected émigrés had large effects.

- **Connects migration, science, and long-run innovation.** The paper bridges economic history, innovation economics, migration, and science policy by showing how displaced human capital can reshape the direction of domestic research.

Incremental contribution in one sentence. Moser, Voena, and Waldinger transform a famous historical episode into a field-level causal analysis showing that displaced high-skilled scientists increased domestic invention primarily by drawing new researchers into their fields and by creating collaborative knowledge networks.

3 Data and measurement

3.1 German and Austrian chemistry professors

- The universe starts from 535 chemistry professors and postdoctoral lecturers at German and Austrian universities.
- Dismissed professors are identified using the List of Displaced German Scholars and additional historical sources.
- Ninety-three chemists were dismissed between 1933 and 1941; about 87 percent were Jewish.
- Twenty-six dismissed chemists who were professionally active in the United States are classified as German Jewish émigrés to the United States.

Interpretation: The historical personnel data define the shock. The paper uses the Nazi dismissals as a plausibly exogenous source of movement in scientific human capital.

3.2 Patent fields and treatment categories

- Research fields are USPTO technology classes.
- The baseline sample contains 166 classes with at least one US patent by an academic chemist from Germany or Austria between 1920 and 1970.
- Sixty classes include at least one patent by an émigré to the United States.
- One hundred six classes include patents by other German chemists but no patents by émigrés.
- For the IV design, forty-eight classes include pre-1933 patents by dismissed chemists, while 118 do not.

Interpretation: Treatment is defined by research field exposure, not by region or firm. This is appropriate because scientific knowledge is organized by fields and patent classes.

3.3 US invention outcome

- The main outcome is the number of US patents by domestic US inventors in class c and year t.
- The patent data cover 1920–1970.
- The aggregate patent-field dataset contains more than 1.3 million US patents by domestic US inventors.
- Citation-weighted and count-data robustness checks are discussed in the paper and online appendix.

Interpretation: Patents measure inventive output in chemistry, where patents are especially important for protecting innovation.

3.4 Inventor-level data

- The authors construct a dataset of all domestic US inventors in the 166 patent classes.
- Incumbents are inventors who patented at least once before 1933.
- Entrants are inventors whose first patent in a field appears after the relevant year.
- The inventor data allow the paper to test whether the main effect comes from incumbent productivity or entry.

Interpretation: The inventor-level data are essential for mechanism. Without them, the paper could only show aggregate field growth.

3.5 Co-inventor and junior émigré data

- Co-inventors are identified from patent documents of senior émigré chemists.
- Second-degree co-inventors are co-inventors of those co-inventors.
- A separate dataset tracks young émigré chemists who were not yet professors in 1933.

Interpretation: These data test whether the effect of émigré professors spread through collaborative networks and whether senior professors proxy for a broader movement.

4 Models and empirical specifications

4.1 Baseline difference-in-differences

The baseline OLS regression is:

$$Patents_{c,t} = \alpha_0 + \beta\, EmigreClass_c \times Post_t + \gamma' X_{c,t} + \delta_t + f_c + \varepsilon_{c,t}.$$

- $EmigreClass_c$ equals one for classes with at least one patent by an émigré to the United States.
- $Post_t$ equals one after the dismissal year: after 1932 for German émigrés and after 1937 for Austrian émigrés if no German émigré is present.
- $X_{c,t}$ includes foreign patents, class age and its square, and patent pool indicators.
- Year fixed effects absorb common shocks to patenting; class fixed effects absorb time-invariant field differences.

Interpretation: The coefficient β measures whether domestic US patenting grows more after the dismissals in fields connected to émigré chemists than in fields of other German chemists.

4.2 Instrumental variables specification

The IV specification instruments $EmigreClass_c \times Post_t$ with:

$$Pre1933DismissedClass_c \times Post_t,$$

where $Pre1933DismissedClass_c$ equals one for patent classes in which dismissed German or Austrian chemists had patented before 1933.

Interpretation: The instrument uses research-field exposure determined before the post-1933 US response. It addresses the concern that émigrés may have chosen fields that were already set to grow in the United States.

4.3 Event-study/year-specific coefficients

The paper also estimates year-specific treatment effects:

$$Patents_{c,t} = \sum_{\tau} \beta_\tau (EmigreClass_c \times 1[t = \tau]) + \gamma' X_{c,t} + \delta_t + f_c + \varepsilon_{c,t}.$$

Interpretation: The year-specific estimates test pretrends and visualize the timing of the effect. A post-1933 increase with no large pretrend supports the causal interpretation.

4.4 Incumbent inventor specification

For incumbent inventors, the paper estimates:

$$Patenting_{i,t} = \alpha + \beta ShareEmigrePatents_i \times Post_t + \gamma' Z_{i,t} + \delta_t + f_i + \varepsilon_{i,t}.$$

Interpretation: If the aggregate increase came from incumbent productivity, β should be positive. Instead, the estimates show no increase and often a decline, implying the main effect is not incumbent productivity.

4.5 Entry specification

For entry into fields, the paper estimates:

$$Entry_{c,t} = \alpha_0 + \beta EmigreClass_c \times Post_t + \gamma' X_{c,t} + \delta_t + f_c + \varepsilon_{c,t}.$$

Interpretation: A positive β means more new inventors enter émigré fields after the dismissals. This is the central mechanism result.

5 Identification and empirical design

5.1 Why the setting is useful

- The Nazi dismissals generated a large, historically sharp displacement of scientists.
- Pre-1933 research fields can be observed before the forced migration and before the main US patenting response.
- Chemistry is a field where patents are a relatively strong measure of innovation.
- The paper observes both aggregate field-level patenting and individual inventor histories.

Interpretation: The setting allows the authors to connect a historical shock to field-level outcomes and then decompose the effect into mechanisms.

5.2 Main threats to identification

- Émigrés may choose fields that would have grown anyway in the United States.
- Fields of émigrés may differ from fields of other German chemists in life-cycle trends.
- US invention may respond to broader scientific shocks unrelated to émigrés.
- Patent counts may not capture all scientific contributions.

Interpretation: The paper addresses these threats with controls for foreign patenting, class age, patent pools, fixed effects, event studies, IV estimates, and mechanism tests.

5.3 Why the IV is informative

- Dismissals were caused by Nazi policy, not by future US research opportunities.
- Pre-1933 fields of dismissed chemists were determined before the migration event.
- First-stage estimates show that pre-1933 dismissed fields strongly predict émigré fields.
- IV estimates exceed OLS estimates, consistent with negative selection of émigrés into less productive US fields.

Interpretation: The IV design is the paper's strongest answer to endogenous field selection.

5.4 Mechanism design

- Incumbent regressions test whether existing US inventors became more productive.
- Entry regressions test whether new inventors entered émigré fields.
- Co-inventor evidence tests whether collaboration networks propagated the effect.
- Young-émigré evidence tests whether senior professors proxy for a broader migration flow.

Interpretation: The paper's contribution is not just estimating an average treatment effect; it decomposes the channel of innovation growth.

6 Section-by-section study guide

- **Introduction.** Presents the historical question and the empirical challenge of immigrant selection.
- **Data.** Constructs the universe of German/Austrian chemists, émigrés, patent classes, domestic US patents, incumbent inventors, entrants, and co-inventors.
- **Aggregate patent effects.** Estimates OLS and IV effects of émigré field exposure on domestic US patenting.
- **Mechanisms.** Uses inventor-level data to distinguish incumbent productivity from entry.
- **Networks.** Studies co-inventors and second-degree co-inventors to understand diffusion.
- **Junior émigrés.** Checks whether fields of senior professors proxy for a broader flow of scientists.
- **Conclusion.** Interprets high-skilled immigration as a source of domestic research entry and long-run innovation.

7 Tables and figures

7.1 Figure 1: US Patents per Year by German Chemists

Read:

- The top panel compares patents by émigrés to the United States with patents by other German chemists.
- The bottom panel compares dismissed chemists with other German chemists.
- Émigré patenting in the United States is low before 1933 and rises after the dismissals, especially in the 1940s and early 1950s.
- Other German chemists patent heavily before the 1940s and decline later as they age.

Detailed interpretation: Figure 1 establishes the historical migration-patenting pattern. It shows that the direct patenting of émigrés in the United States grows only after the Nazi dismissals, consistent with the displacement shock. It also shows why age and timing matter: other German chemists have a different life-cycle profile.

Economic message: The figure motivates the central question: did émigrés merely patent themselves, or did they change the direction and productivity of domestic US innovation?

7.2 Table 1: Summary Statistics – US Patents by Domestic Inventors across USPTO Classes

Read:

- Classes with émigré patents nearly double in domestic US patents after 1933, rising from about 149 to 287 patents per class-year.
- Classes with other German chemists rise much less, from about 218 to 249 patents per class-year.
- Classes with more émigré patents show larger post-1933 growth.
- The IV-relevant classes with pre-1933 patents of dismissed chemists are also summarized.

Detailed interpretation: Table 1 is the descriptive DiD table. It reveals that émigré fields were initially less active in domestic US patenting, but grew much faster after 1933. This baseline pattern is consistent with both historical accounts and the negative-selection interpretation: émigrés may have entered fields where preexisting US patenting was weaker.

Economic message: The descriptive statistics already show a large relative increase in US invention in émigré fields.

7.3 Figures 2A and 2B: US Patents per Class and Year by Domestic US Inventors

Read:

- Figure 2A compares research fields of émigrés with fields of other German chemists.
- Figure 2B splits émigré fields by the number of émigré patents.
- Domestic patenting rises more sharply after 1933 in émigré fields, especially fields with more émigré patents.

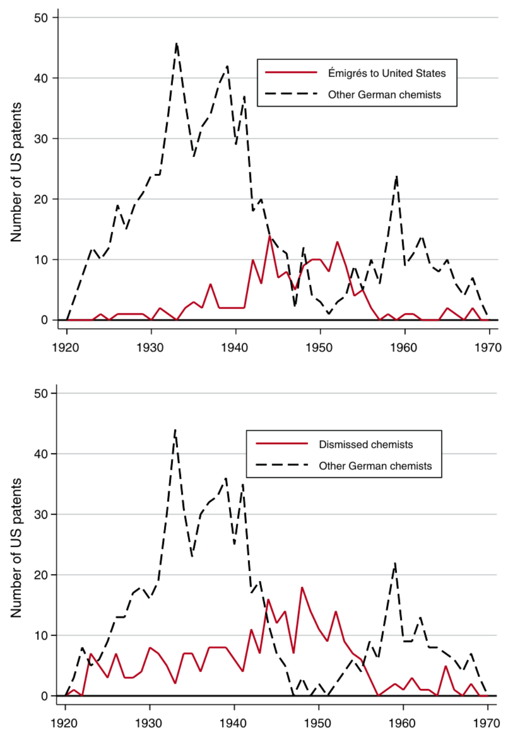

FIGURE 1. US PATENTS PER YEAR BY GERMAN CHEMISTS

Data cover 946 US patents by 535 professors and lecturers of chemistry at German and Austrian universi
)33 is the year of the first dismissals. The top panel shows patent issues per year for chemists who emigrate

Figure 1: US Patents per Year by German Chemists.

TABLE 1—SUMMARY STATISTICS: US PATENTS BY DOMESTIC INVENTORS ACROSS USPTO CLASSES

	All classes	Classes with 1920–1970 patents by US émigrés	Classes without 1920–1970 patents by US émigrés	Classes with pre-1933 patents by dismissed	Classes without pre-1933 patents by dismissed
	(1)	(2)	(3)	(4)	(5)
Patents by US inventors 1920–1970	2,073,771	771,377	1,302,394	619,308	1,454,463
Number of classes	166	60	106	48	118
Mean class age in 1932	87.2	84.6	88.7	87.4	87.3
p-value of equality of means test			0.085		0.929
Mean number of foreign patents in 1932	0.93	0.92	0.93	0.70	1.01
p-value of equality of means test			0.942		0.216
Mean patents per class and year 1920–1970	244.95	252.08	240.92	252.99	241.69
Mean patents per class and year 1920–1932	193.39	149.25	218.38	157.50	207.99
Mean patents per class and year 1933–1970	262.59	287.26	248.63	285.65	253.21

Notes: Data include patent-main class combinations of US inventors in classes with 1920–1970 patents by German university chemists. Patents by US inventors in these classes were collected from www.uspto.gov. Dismissed and

Table 1: Summary Statistics – US Patents by Domestic Inventors across USPTO Classes.

Detailed interpretation: These figures visually motivate the regression estimates. Figure 2A shows the main treatment-control comparison. Figure 2B provides an intensity check: fields with more émigré activity experience larger post-1933 increases, which strengthens the interpretation that exposure matters.

Economic message: The aggregate effect is not confined to one or two fields; it increases with treatment intensity.

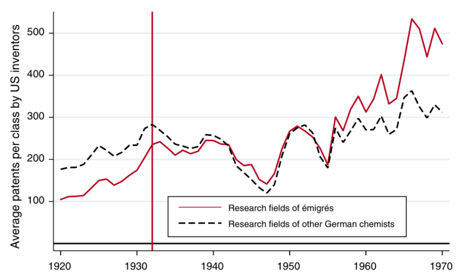

FIGURE 2A. US PATENTS PER CLASS AND YEAR BY DOMESTIC US INVENTORS IN RESEARCH FIELDS OF ÉMIGRÉS AND OTHER GERMAN CHEMISTS

Notes: Data cover 2,073,771 patent-main class combinations by US inventors across 166 research fields defined at the level of USPTO classes. *Research fields of émigrés* cover 60 classes which include at least one patent between 1920 and 1970 by a German or Austrian émi-

(a) Figure 2A

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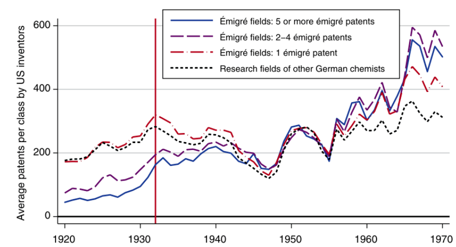

FIGURE 2B. US PATENTS PER CLASS AND YEAR BY DOMESTIC US INVENTORS IN RESEARCH FIELDS OF ÉMIGRÉS AND OTHER GERMAN CHEMISTS

Notes: Data cover 2,073,771 patent-main class combinations across 166 research fields defined at the level of USPTO classes. *Émigré fields: 5 or more émigré patents* include classes which include 5 or more patents between 1920 and 1970 by German or Austrian émigrés to the United

(b) Figure 2B

7.4 Table 2 and Figure 3: OLS Estimates of Changes in US Patents

Read:

- Table 2 reports OLS DiD estimates. The baseline treatment effect is 105.2 additional patents per year in émigré fields.
- Controls for foreign patents, class age, and patent pools reduce the coefficient but it remains positive and significant.
- Figure 3 reports year-specific estimates, showing the effect grows after the dismissals.

Detailed interpretation: Table 2 is the main baseline regression table. It establishes that the descriptive pattern survives fixed effects and controls. Figure 3 tests timing: the post-1933 increase appears after the émigrés' arrival, which supports the causal timing.

Economic message: The OLS evidence indicates a large field-level increase in domestic US innovation after émigrés arrived.

TABLE 2—ORDINARY LEAST SQUARES REGRESSIONS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Émigré class × post	105.222***	91.712***	84.803***	75.439***				
	(22.203)	(19.212)	(18.950)	(19.326)				
Number émigré patents × post					5.848*	4.992*	4.527**	3.991**
					(3.058)	(2.561)	(2.182)	(1.956)
Number foreign patents	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Quadratic class age	No	No	Yes	Yes	No	No	Yes	Yes
Patent pools	No	No	No	Yes	No	No	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,466	8,466	8,466	8,466	8,466	8,466	8,466	8,466
R^2	0.783	0.845	0.849	0.851	0.779	0.842	0.846	0.848

Notes: The dependent variable is patents by US inventors per USPTO class and year, excluding patents by émigrés. *Émigré class* equals 1 for classes which include at least one US patent by an émigré. *Number émigré patents* measures the number of US patents by émigrés in class c. Classes without émigré patents form the control group. The dummy variable *Post* equals 1 for years after the dismissals. *Number of foreign patents* counts US patents by foreign nationals in class c and year t. *Quadratic class age* is a second-degree polynomial for years since the first patent in class c. The indicator variable *patent pools* equals 1 for classes which were affected by a patent pool.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

(a) Table 2: OLS regressions

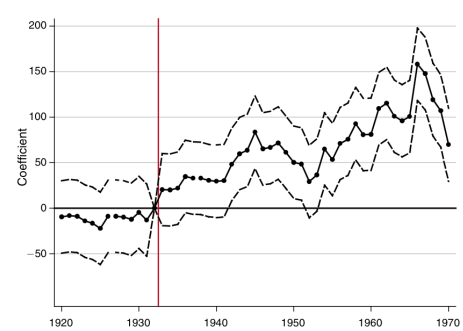

FIGURE 3. YEAR-SPECIFIC OLS ESTIMATES US PATENTS PER YEAR IN RESEARCH FIELDS OF ÉMIGRÉS

(b) Figure 3: year-specific OLS estimates

7.5 Table 3, Figure 4, and Table 4: Instrumental-Variables Evidence

Read:

- Table 3 reports first-stage and reduced-form evidence using pre-1933 fields of dismissed chemists.
- Figure 4 shows that fields with pre-1933 patents by dismissed chemists grow relative to fields of other German chemists after 1933.
- Table 4 reports IV estimates; the IV coefficients are larger than OLS estimates.

Detailed interpretation: The IV results address endogenous field choice. If émigrés had selected fields that were already poised to grow, the IV estimate should be smaller or unstable. Instead, IV is larger, consistent with historical evidence that émigrés to the United States were often negatively selected into less productive or less established fields.

Economic message: The IV evidence suggests the OLS estimates understate the effect of émigrés on US invention.

TABLE 3—FIRST STAGE AND REDUCED FORM

	First stage							
	Émigré class × *post*		Number *émigré patents* × *post*		Reduced form			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dismissed class × *post*	0.370***	0.339***			80.821***	57.752***		
	(0.081)	(0.079)			(23.155)	(19.436)		
Number dismissed patents × *post*			1.384***	1.303***			35.595***	22.330***
			(0.442)	(0.435)			(6.547)	(6.339)
Number foreign patents	No	Yes	No	Yes	No	Yes	No	Yes
Quadratic class age	No	Yes	No	Yes	No	Yes	No	Yes
Patent pools	No	Yes	No	Yes	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,466	8,466	8,466	8,466	8,466	8,466	8,466	8,466
R^2	0.801	0.809	0.770	0.773	0.779	0.849	0.782	0.849
F-statistic	20.80	18.25	9.79	8.99				

Notes: Dependent variables are *émigré class* × *post* (columns 1–2), *number of émigré patents* × *post* (columns 3–4), and *patents per class* and *year by US inventors* (columns 5–8). In first-stage regressions (columns 1–4), the dependent variables are *Émigré class* × *post* (columns 1 and 2) and *Number émigré patents* × *post* (columns 3 and 4). *Émigré class* equals 1 for classes which include at least one US patent by an émigré. *Number émigré patents* measures the number of US patents by émigrés in class *c*. *Dismissed class* equals 1 for classes which include at least one pre-1933 US patent by a dismissed chemist. *Number dismissed patents* indicates the number of pre-1933 US patents by dismissed chemists in each class. The dummy variable *Post* equals 1 for years after the dismiss-

(a) Table 3: first stage and reduced form

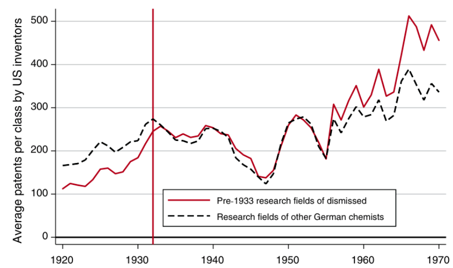

FIGURE 4. PATENTS BY DOMESTIC INVENTORS IN RESEARCH FIELDS IN WHICH DISMISSED CHEMISTS WERE ACTIVE BEFORE 1933

Notes: Data cover 2,073,771 patent-main class combinations by US inventors across 166 research fields defined at the level of USPTO classes. *Pre-1933 research fields of dismissed*

(b) Figure 4: dismissed-field patenting

TABLE 4—INSTRUMENTAL VARIABLE REGRESSIONS

	(1)	(2)	(3)	(4)
Émigré class × *post*	218.707***	170.136***		
	(60.614)	(57.992)		
Number émigré patents × *post*			25.717***	17.137**
			(8.750)	(6.909)
Number foreign patents	No	Yes	No	Yes
Quadratic class age	No	Yes	No	Yes
Patent pools	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Class fixed effects	Yes	Yes	Yes	Yes
Observations	8,466	8,466	8,466	8,466

Notes: The dependent variable is patents by US inventors per USPTO class and year, excluding US patents by émigrés. *Émigré class* equals 1 for classes that include at least one US patent by an émigré. *Number émigré patents* measures the number of US patents by émigrés in class *c*. Classes without émigré patents form the control. The dummy variable *Post* equals 1 for years after the dismissals. Instruments are *Dismissed class* × *post* (columns 1 and 2) and *number dismissed patents* × *post* (columns 3 and 4). *Dismissed class* equals 1 for classes that include at least one pre-1933 US patent by a dismissed chemist. *Number dismissed patents* indicates the number of pre-1933 US patents by dismissed chemists in each class. *Number of foreign patents*

Table 4: Instrumental-variable regressions.

7.6 Tables 5–7 and Figures 5–6: Incumbent Inventors

Read:

- Table 5 describes 210,410 US inventors active before 1933.
- Table 6 shows that incumbents more exposed to émigré fields do not become more likely to patent after 1933; with controls, effects are negative.
- Table 7 confirms the first-stage/reduced-form logic for incumbent exposure.
- Figures 5 and 6 show no differential rise in incumbent patenting after 1933.

Detailed interpretation: This block is crucial for mechanism. If émigrés increased innovation by raising incumbent productivity, then incumbents already active in émigré fields should become more productive. Instead, the data show no such increase. This shifts the mechanism toward entry by new inventors.

Economic message: The innovation boom did not primarily come from incumbents becoming more productive; it came from changes in who entered the fields.

TABLE 5—SUMMARY STATISTICS: US PATENTS BY DOMESTIC INVENTORS WHO WERE ACTIVE PRIOR TO 1933

	All inventors	Fraction of patents in research fields of émigrés			Fraction of pre-1933 patents in research fields of dismissed chemists		
		< 50%	50%	> 50%	< 50%	50%	> 50%
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Total inventors active before 1933	210,410	144,647	7,842	57,921	155,261	4,719	50,430
Annual probability of patenting 1920–1970	0.035	0.034	0.050	0.036	0.035	0.067	0.034
Annual probability of patenting 1920–1932	0.098	0.098	0.120	0.097	0.098	0.161	0.094
Annual probability of patenting 1933–1970	0.014	0.013	0.026	0.015	0.013	0.036	0.013
Patents per inventor and year 1920–1970	0.043	0.042	0.055	0.045	0.043	0.084	0.042
Patents per inventor and year 1920–1932	0.112	0.111	0.132	0.111	0.111	0.184	0.107
Patents per inventor and year 1933–1970	0.020	0.018	0.029	0.023	0.019	0.050	0.019

Notes: Data include 210,410 US patentees with at least one patent between 1920 and 1932. We constructed data on patents per year of these patentees through a search algorithm, which identified patents by individual inventors per class and year, using Google's *Patent Grant Optical Character Recognition* (*OCR*) *Text* (*1920–1979*) database. The

(a) Table 5: incumbent summary statistics

TABLE 6—ORDINARY LEAST SQUARES AND INSTRUMENTAL VARIABLES

	OLS (linear probability)		IV	
	(1)	(2)	(3)	(4)
Share of patents in émigré classes × post	0.002**	−0.007***	−0.001	−0.022***
	(0.001)	(0.002)	(0.002)	(0.006)
Quadratic time to first patent	No	Yes	No	Yes
Quadratic time since first patent	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Inventor fixed effects	Yes	Yes	Yes	Yes
Observations	10,730,910	10,730,910	10,730,910	10,730,910
R^2	0.045	0.147	—	—

Notes: Dependent variable is patenting by US inventors. The dependent variable equals one if inventor i obtains at least one patent in year t, and 0 otherwise. The sample includes all domestic US patentees with at least one patent between 1920 and 1932. *Share of patents in émigré classes* measures the total share of patents by a US inventor that are in the 60 research fields of émigrés. The variable *Post* equals 1 for years after the dismissals. *Quadratic time to first patent* is a second-degree polynomial for years until an inventor patents for the first time in any of the 166 classes. *Quadratic time since first patent* is a second-degree polynomial for years after an inventor patents for the first time in any of the 166 classes.

*** Significant at the 1 percent level.
** Significant at the 5 percent level.
* Significant at the 10 percent level.

(b) Table 6: incumbent OLS and IV

TABLE 7—FIRST STAGE AND REDUCED FORM

	First stage		Reduced form	
	(1)	(2)	(3)	(4)
Share of pre-1933 patents in dismissed classes × post	0.403***	0.402***	−0.0003	−0.009***
	(0.086)	(0.086)	(0.001)	(0.002)
Quadratic time to first patent	No	Yes	No	Yes
Quadratic time since first patent	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Inventor fixed effects	Yes	Yes	Yes	Yes
Observations	10,730,910	10,730,910	10,730,910	10,730,910
F-statistic	21.83	21.65		
R^2	0.434	0.434	0.045	0.147

Notes: In columns 1–2 the dependent variable *Share of patents in émigré classes × post* measures the total share of patents by a US inventor that are in the 60 research fields of émigrés. In colunns 3–4, the dependent variable equals 1 if inventor i obtains at least one patent in year t, and 0 otherwise. *Share of pre-1933 patents in dismissed classes* measures the share of a domestic US inventor's pre-1933 patents that are in 48 classes with pre-1933 patents of dismissed chemists.

*** Significant at the 1 percent level.

(a) Table 7: incumbent first stage/reduced form

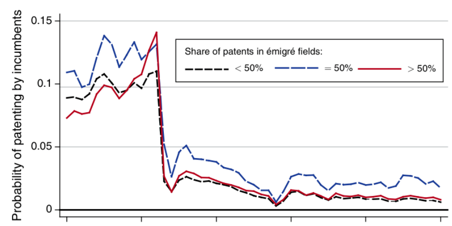

FIGURE 5. PATENTING PER YEAR BY INCUMBENT INVENTORS IN RESEARCH FIELDS OF ÉMIGRÉS COMPARED WITH FIELDS OF OTHER GERMAN CHEMISTS

Notes: Probability of patenting by incumbents measures the average probability of patenting per year by 210,410 inventors who patented at least one invention before 1933. *Share of patents in émigré fields* measures the share of all patents (1920–1970) by an individual inventor that are in a class with at least one patent by an émigré.

(b) Figure 5: incumbents in émigré fields

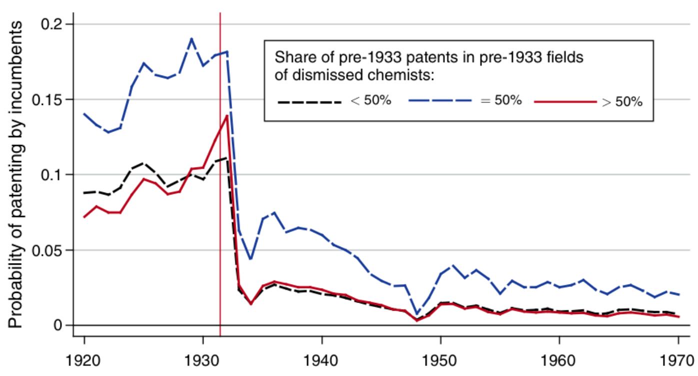

FIGURE 6. PATENTING PER YEAR BY INCUMBENT INVENTORS IN PRE-1932 FIELDS OF DISMISSED CHEMISTS COMPARED WITH FIELDS OF OTHER GERMAN CHEMISTS

Notes: Probability of patenting by incumbents measures the average probability of patent-

Figure 6: incumbents in pre-1933 dismissed fields.

7.7 Table 8, Figure 7, Table 9, and Figure 8: Entry of New Patentees

Read:

- Table 8 shows that entry into émigré fields increases after 1933, especially among inventors without prior patents in the 166 classes.
- Figure 7 shows entry into émigré fields rising after the dismissals.
- Table 9 estimates that émigré fields gained substantial additional entrants per class-year; IV estimates are larger than OLS.
- Figure 8 confirms similar entry patterns using pre-1933 fields of dismissed chemists.

Detailed interpretation: Entry is the paper's main mechanism result. The aggregate increase in patents is explained by more US researchers entering émigré-related fields, not by incumbent inventors suddenly patenting more. The distinction between entrants into fields and entrants into patenting helps show that the effect is not only a reshuffling of established inventors across related classes.

Economic message: Émigrés changed the direction of domestic research by attracting new US inventors to their fields.

TABLE 8—SUMMARY STATISTICS ON ENTRY OF NEW PATENTEES ACROSS RESEARCH FIELDS

	All classes (1)	Classes with 1920–1970 patents by US émigrés (2)	Classes w/o 1920–1970 patents by US émigrés (3)	Classes with pre-33 patents by dismissed (4)	Classes w/o pre-33 patents by dismissed (5)
Number of classes	166	60	106	48	118
Panel A. Entrants into research fields					
Total entrants into classes 1920–1970	1,396,318	499,417	896,901	404,927	991,391
Mean entrants per class and year 1920–1970	164.9	163.2	165.9	165.4	164.7
Mean entrants per class and year 1920–1932	153.8	116.1	175.1	121.6	166.8
Mean entrants per class and year 1933–1970	168.8	179.3	162.8	180.4	164.0
Panel B. Entrants into patenting					
Total entrants (no prior patents) 1920–1970	964,526	327,224	637,302	268,084	696,442
Mean entrants (no prior patents) per class and year 1920–1970	113.9	106.9	117.9	109.5	115.7
Mean entrants (no prior patents) per class and year 1920–1932	125.0	92.0	143.8	97.2	136.3
Mean entrants (no prior patents) per class and year 1933–1970	110.1	112.1	109.0	113.7	108.7

Notes: Entrants are patentees who patent for the first time in 1 of 166 research fields, defined at the level of USPTO

(a) Table 8: entry summary statistics

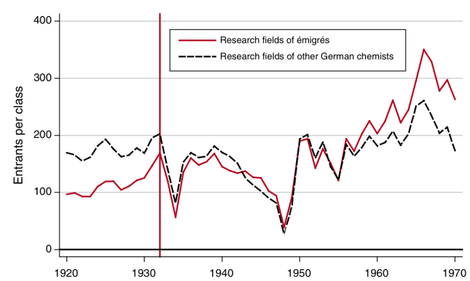

FIGURE 7. ENTRY OF US PATENTEES INTO RESEARCH FIELDS OF ÉMIGRÉS COMPARED WITH FIELDS OF OTHER GERMAN CHEMISTS

Notes: Entrants per class measures the number of new researchers who entered the average

(b) Figure 7: entry into émigré fields

TABLE 9—ORDINARY LEAST SQUARES AND INSTRUMENTAL VARIABLES REGRESSIONS

	OLS				Instrumental Variables			
	Entrants into field		Entrants into patenting		Entrants into field		Entrants into patenting	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Émigré class × post	73.799***	58.181***	53.434***	43.967***	162.287***	142.119***	116.707***	109.466***
	(15.674)	(14.715)	(12.522)	(12.261)	(44.195)	(45.982)	(34.565)	(37.863)
Number foreign patents	No	Yes	No	Yes	No	Yes	No	Yes
Quadratic class age	No	Yes	No	Yes	No	Yes	No	Yes
Patent pools	No	Yes	No	Yes	No	Yes	No	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Class fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,466	8,466	8,466	8,466	8,466	8,466	8,466	8,466
R^2	0.781	0.835	0.763	0.805	0.767	0.824	0.750	0.792

Notes: In columns 1–2 and 5–6, the dependent variable is number of new patentees per year in class c without prior patents in class c. In columns 3–4 and 7–8, the dependent variable is number of new patentees per year in class c with neither prior patents in class c nor prior patents in any other of the 166 classes. *Émigré class* equals 1 for classes that include at least one US patent by an émigré. Classes without émigré patents form the control group. We instrument with *Pre-1933 Dismissed class × Post* for *Émigré class × post*. *Pre-1933 Dismissed class* equals 1 for classes that include at least one pre-1933 US patent by a dismissed chemist. The dummy variable *Post* equals 1 for years after the dismissals. First stage regressions are reported in column 2 of Table 3. *Number of foreign patents* counts US patents by foreign nationals in class c and year t. *Quadratic class age* is a second-degree polynomial for years since the first patent in class c. The indicator variable *patent pools* equals 1 for classes that were affected by a patent pool.

(a) Table 9: entry regressions

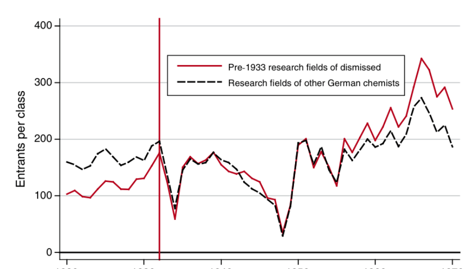

FIGURE 8. ENTRY OF US PATENTEES INTO RESEARCH FIELDS WITH PRE-1933 PATENTS OF DISMISSED COMPARED WITH RESEARCH FIELDS OF OTHER GERMAN CHEMISTS

Notes: Entrants per class measures the number of new researchers who entered the average research field in year t. Entry into a research field is defined by the first patent of an inventor in a USPTO patent class. *Pre-1933 research fields of dismissed* consist of 48 USPTO classes that

(b) Figure 8: entry into dismissed fields

7.8 Table 10 and Figure 9: Co-Inventor Networks

Read:

- Table 10 shows that co-inventors and co-inventors of co-inventors patent disproportionately in émigré fields after 1933.
- Before 1933, co-inventor activity in émigré fields is minimal; after 1933, it rises sharply.
- Figure 9 shows co-inventors' patents in émigré fields increasing especially after 1940 and remaining high through the 1950s.

Detailed interpretation: The co-inventor evidence identifies a diffusion channel. Émigré professors did not only contribute directly; they appear to have trained or collaborated with US scientists who continued to innovate in the same fields. The lag in Figure 9 is consistent with knowledge transmission taking time.

Economic message: Scientific migration can generate persistent innovation through collaboration networks and training effects.

TABLE 10—PATENTING OF CO-INVENTORS, AND CO-INVENTORS OF CO-INVENTORS OF ÉMIGRÉ CHEMISTS

	Patents in 166 technology classes		
	Patents assigned only to 60 émigré classes	Patents assigned only to 106 non-émigré classes	Patents assigned to both émigré and non-émigré classes
Panel A. Co-inventors of senior émigrés			
1920–1932	4	4	0
1933–1970	469	24	75
1920–1970	473	28	75
Panel B. Co-inventors of co-inventors of senior émigrés			
1920–1932	48	59	24
1933–1970	1,103	162	264
1920–1970	1,151	221	288

Notes: Data for panel A include include co-inventors of senior émigrés; which we identified from the list of inventors on patent grants. Data for panel B include co-inventors of co-inventors (second degree co-inventors) of senior émigrés. Data on 1920–1970 patents of co-inventors were hand-collected from Google Patents (www.patents.google.com). Data on 1920–1970 patents of co-inventors of co-inventors were collected with an algorithm using the inventor data.

(a) Table 10: co-inventors

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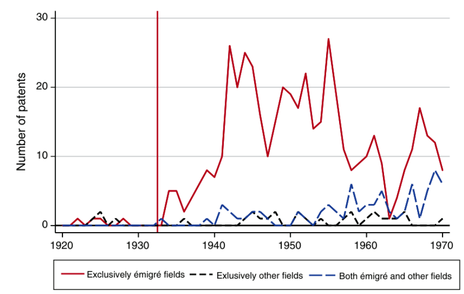

FIGURE 9. PATENTS BY CO-INVENTORS OF ÉMIGRÉS

Notes: Total patents by 47 co-inventors of émigrés. *Exclusively émigré fields* measures the

(b) Figure 9: co-inventor patents

7.9 Table 11: Patenting of Young Émigré Chemists

Read:

- Junior émigré chemists produced 175 patents in the 166 technology classes.
- After 1933, 113 patents were assigned only to senior-émigré classes, 22 only to non-émigré classes, and 34 to both.
- The concentration of junior émigré patents in senior-professor fields suggests that senior professor fields proxy for a broader German Jewish scientific migration.

Detailed interpretation: Table 11 addresses external validity. The main analysis observes prominent senior professors, but the broader migration included younger scientists. The table shows that junior émigrés were active in similar fields, suggesting the main treatment categories capture more than the visible senior cohort.

Economic message: The paper's measured senior-emigrant fields may proxy for a broader flow of displaced German Jewish scientific human capital.

TABLE 11—PATENTING OF YOUNG ÉMIGRÉ CHEMISTS

	Patents in 166 technology classes		
	Patents assigned only to 60 émigré classes	Patents assigned only to 106 non-émigré classes	Patents assigned to both émigré and non-émigré classes
1920–1932	6	0	0
1933–1970	113	22	34
1920–1970	119	22	34

Notes: Data include young émigré chemists as listed in Strauss et al. (1983). Data on 1920–1970 patents were collected from Google Patents (www.patents.google.com).

Table 11: Patenting of Young Émigré Chemists.

8 Result map

- **Direct émigré patenting:** émigré patenting in the United States rises after 1933.
- **Aggregate field effect:** domestic US patents rise more in émigré fields than in fields of other German chemists.
- **IV evidence:** pre-1933 fields of dismissed chemists strongly predict émigré fields and imply larger effects than OLS.
- **Incumbent mechanism rejected:** incumbents in émigré fields do not become more productive.
- **Entry mechanism supported:** new US inventors enter émigré fields after 1933.
- **Network channel:** co-inventors and second-degree co-inventors become especially active in émigré fields.
- **External validity check:** younger émigré chemists also patent mainly in senior-emigrant fields.

9 Short summary

- The paper studies whether German Jewish émigré chemists increased US innovation.
- It uses patents by domestic US inventors in 166 USPTO technology classes between 1920 and 1970.
- OLS DiD estimates imply a substantial increase in patenting in émigré fields after 1933.
- IV estimates using pre-1933 fields of dismissed chemists are even larger.
- The effect is not driven by incumbent US inventors becoming more productive.
- The effect is driven mainly by entry of new US inventors into émigré fields.
- Co-inventor networks amplified and prolonged the effect.
- The fields of senior émigré professors also capture the activity of younger émigré chemists.
- The key lesson is that high-skilled immigrants can transform innovation by redirecting domestic talent into new research fields.

10 Conclusion

10.1 Core conclusion

Moser, Voena, and Waldinger show that German Jewish émigré chemists caused a substantial increase in US invention in their research fields. The effect is visible in aggregate patenting, stronger in instrumental-variable estimates, and concentrated in entry by new domestic inventors rather than productivity gains by incumbents.

10.2 Economic intuition

High-skilled migrants can do more than produce their own inventions. They can redirect research agendas, attract new domestic researchers, and build collaboration networks. The émigrés' arrival changed the relative attractiveness of their research fields and encouraged US inventors to enter those areas.

10.3 Incremental contribution revisited

The paper's incremental contribution is to identify a causal field-level effect of displaced scientific human capital and to decompose the mechanism. It shows that high-skilled immigration can reshape the direction of domestic innovation by attracting new entrants and building collaborative networks.

10.4 Policy implication

The findings support the idea that admitting high-skilled scientists can have innovation benefits beyond their direct output. The effect operates through training, networks, and field entry, which may unfold over decades.

10.5 Limitations

The analysis focuses on chemistry and patented inventions, so it may miss nonpatented or classified scientific contributions. The main treatment group consists of prominent senior professors, though the junior-emigrant evidence suggests their fields proxy for a broader movement. The IV design addresses field selection but still relies on the assumption that pre-1933 dismissed fields affected post-1933 US patenting only through émigré-related exposure.

10.6 Takeaway

The paper's core lesson is that the movement of scientific talent can have large and persistent effects on innovation, especially when migrants draw new researchers into their fields and transmit knowledge through collaboration networks.